# Collections & Sorting

covering

** Java Collections Framework

** **`Comparable`** interface

** **`ArrayList`** // 2D arrays

** (Brief introduction to) Sorting Algorithms:
*Bubble Sort*, *Insertion Sort*

Chapter 9 (sections 9.1-9.4, 9.6) – “Core Java” book
Chapter 11– “Head First Java” book
Chapters 6, 25 (sections 6.10-6.11, 25.1-25.4) – “Introduction to Java Programming” book
Chapter 8 – “Java in a Nutshell” book

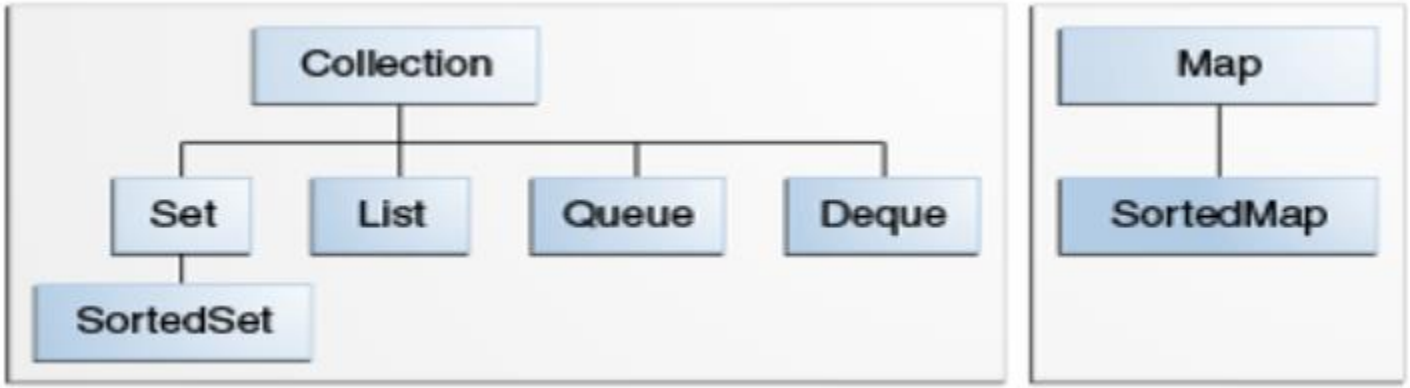
# Java's Collection Classes

- Most programs utilise *collections of data* e.g.,
  - a set of users;
  - words in a dictionary.

Example of another data structure?

- Collection in Java: "an object that groups multiple elements into a single unit".
  - Can be used to: store, retrieve, manipulate data.
  - Examples: a poker hand, a mail folder, a telephone directory.
- Java provides several interfaces, implementations and algorithms for handling collections of objects, via its Java Collections Framework (see `java.util` package).

# Collection Framework: Interfaces




- Set: A collection that contains no duplicate elements; it models the mathematical *set* abstraction.
- List: An ordered collection (also known as a *sequence*). Elements can be accessed by their position in the list, and it is possible to search for elements in the list. Lists allow for duplicate elements.
- Map: An object that maps keys to values. A map does not contain duplicate keys; each key can map to at most one value.

See descriptions of the other interfaces in the Java API.

# Choosing a Collection …

- Which collection you choose will depend on many factors e.g.,
  - whether the collection needs to be of fixed size *or* dynamic;
  - whether it has a natural order *or* not;
  - whether we want to insert/delete at arbitrary places, *or* only at the start or end;
  - whether we want to be able to search through a large amount of data very quickly;
  - whether we want random access to elements *or* sequential access;
  - whether we need to maintain key/value pairs.

# Collection Framework: Implementations

| Interfaces | Hash table Implementations | Resizable array Implementations | Tree Implementations | Linked list Implementations | Hash table + Linked list Implementations |
|---|---|---|---|---|---|
| Set | HashSet | | TreeSet | | LinkedHashSet |
| List | | ArrayList | | LinkedList | |
| Queue | | | | | |
| Deque | | ArrayDeque | | LinkedList | |
| Map | HashMap | | TreeMap | | LinkedHashMap |

An array that can grow or shrink.

- Array *versus* **`ArrayList`**:
  - An array needs to know its size at time of creation, but an **`ArrayList`** does not,

```
String[] myStringArray = new String[6];
ArrayList<String> myArrayList = new ArrayList<String>();
```

  - To assign an object in a regular array, you must assign it to a specific index, but with an **`ArrayList`** you needn't,

```
anotherList[4] = b;                anotherArrayList.add(b);
```

# Map and HashMap

- The **Map** interface associates keys (which are like *indexes*) to elements.
  - It cannot contain duplicate keys.
  - Each key maps to one value only.
- The **HashMap** class implements **Map** and is efficient for locating a value, as well as inserting and deleting a mapping.
  - Entries are not ordered.
- Example:

```
import java.util.*;
public class HashMapTester {
   public static void main( String[] args ) {
      Map<String, String> petSounds = new HashMap<String, String>();
      petSounds.put("cat", "Meow");  petSounds.put("mouse", "Squeak");
      petSounds.put("dog", "Woof");  petSounds.put("guineaPig", "Squeak");
      System.out.println("map = " + petSounds);
      String val = (String)petSounds.get("dog");
      System.out.println("Value for key 'dog' is: " + val);
   }
}
```

Output …

```
> java HashMapTester
map = {mouse=Squeak, cat=Meow,
       guineaPig=Squeak, dog=Woof}
Value for key 'dog' is: Woof
```

# Enumerations

- Enumerations are a Java utility class for holding lists of objects.
  - The idea is that we often encounter pseudo-code design like this:

```
lst is a list of objects
for each item i in lst do {
  process i
}
```

  - Enumerations allow us to code this logic directly.

**`Enumeration`** interface:

a) introduced before Java 2;
b) *superseeded* by the **`Iterator`** interface.

# Example using the `Iterator` Interface

**`Iterator`** methods …

| Result | Method | Description |
|---|---|---|
| b = | *it*.**hasNext**() | `true` if there are more elements for the iterator. |
| obj = | *it*.**next**() | Returns the next object. If a generic list is being accessed, the iterator will return something of the list's type. Pre-generic Java iterators always returned type `Object`, so a downcast was usually required. |
| | *it*.remove() | Removes the most recent element that was returned by `next`. Not all collections support `delete`. An *UnsupportedOperationException* will be thrown if the collection does not support *remove()*. |

```
ArrayList<String> alist = new ArrayList<String>();
// Add Strings to alist

for (Iterator<String> it = alist.iterator(); it.hasNext(); ) {
  String s = it.next(); // No downcasting required.
  System.out.println(s);
}
```

… and things for you to try out!

# 2-dimensional (2D) Arrays

- Java stores a 2D array as an array of arrays, e.g.

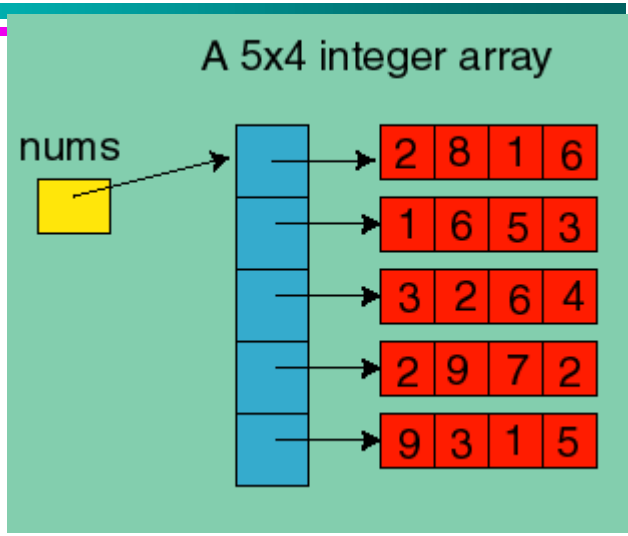
```
int[][] nums = new int[5][4];
```

- When declaring a 2D array:
  - must *always* specify the first dimension

```
nums = new int[][];    // ILLEGAL
```

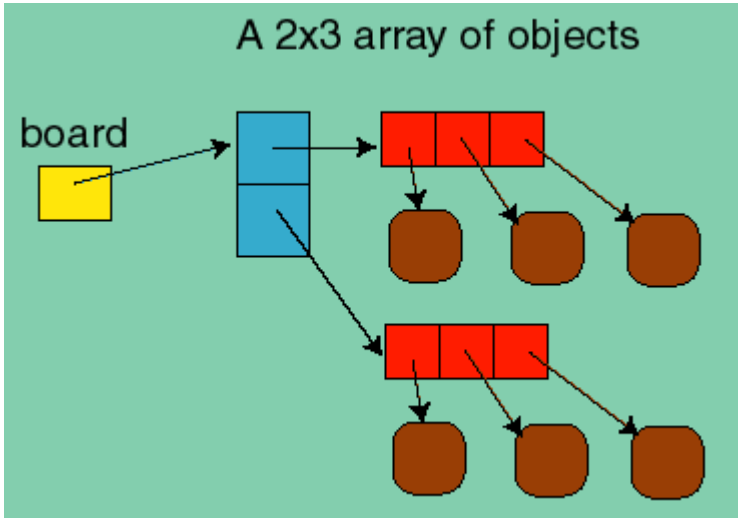
  - do *not* need to specify the second dimension

```
nums = new int[5][];   // OK
nums = new int[5][4]; // OK
```




- A 2D array of objects is an array of an array of references to objects, e.g.

```
Square[][] board = new Square[2][3];
```



Arrays are not part of the Collections framework.

# Practice Exercise 1 – 2D arrays

- What is the output?

```
public class TwoDArrayTester {
  public static void main(String[] args) {
    char[][] pic = new char[6][6];
    for (int i = 0; i < 6; i++) {
      for (int j = 0; j < 6; j++) {
        if ((i == j) || (i == 0) || (i == 5))
         pic[i][j] = '*';
        else pic[i][j] = '.';
      }
    }
    for (int i = 0; i < 6; i++) {
      for (int j = 0; j < 6; j++)
        System.out.print(pic[i][j]);
      System.out.println();
    }
  }
}
```

Output …

# Practice Exercise 2 – 2D ArrayList

- What is the output?

```
import java.util.ArrayList;
public class TwoDArrayListTester {
  public static void main(String[] args) {
    ArrayList<ArrayList<Integer>> topList =
                                  new ArrayList<ArrayList<Integer>>();
    for (int i = 0; i < 3; i++)
      topList.add(new ArrayList<Integer>());

    for (int i = 0; i < 3; i++) {
      for (int j = 0; j < 3; j++)
        topList.get(i).add(new Integer(i+j));
    }
    for (int i = 0; i < 3; i++) {
      for (int j = 0; j < 3; j++)
        System.out.print(topList.get(i).get(j) + " ");
      System.out.println();
    }
  }
}
```

Output …

# Sorting in Java

- Sorting: Common task when programming, e.g. displaying the words in a list in alphabetical order.
  - Algorithms for Sorting:
    - ***Selection*** sort
      1. Find largest number and put it in the last position.
      2. Find next largest number and put it next to last one.
      3. Repeat until finished.
    - ***Insertion*** sort: List of values is sorted by inserting (repeatedly) an unsorted element into a sorted sublist until the complete list is sorted.
    - ***Bubble*** sort (***next*** *…*)

See https://www.cs.usfca.edu/~galles/visualization/ComparisonSort.html for more sorting algorithms (with animations).

# Bubble (or *Sinking*) Sort

- **Bubble** sort:
  - Several passes are made through the array.
  - Each time, successive adjacent pairs are compared.
    - If pair is in decreasing order, order of values is swapped.
    - Otherwise, move on to next pair.

```
boolean changed = true;
do {
  changed = false;
  for (int j=0; j < list.length-1; j++) {
    if (list[j] > list[j+1]) {
      swap list[j] with list[j+1];
      changed = true;
    }
  }
} while (changed);
```

# Practice Exercise 3

- Write a sort method that uses the *bubble-sort algorithm*.
  - Use `{5.0, 4.4, 1.9, 2.9, 3.4, 2.9, 3.5}` to test the method.

Homework: Try doing Programming Exercise 6.17 (IJP book).

# Comparing objects … (1/2)

- Real objects are often “comparable”, e.g. you (students) are much younger than me! ☺
- When writing OO programs, it may be necessary to compare instances of the same class; once instances are comparable, they can be sorted e.g.
  - Given two **`Employee`** objects, it may be necessary to know which one has been with the company for longer.

---

- Java defines two ways of comparing objects:
  1. The objects implement the **`Comparable`** interface.
  2. A **`Comparator`** object is used to compare the two objects.
- If the objects are **`Comparable`**, they are said to be sorted by their "natural" order.
- A **`Comparable`** object can only offer one form of sorting.
  - To provide multiple forms of sorting, **`Comparators`** must be used.

**`Comparator`** is *out of scope* in this course.

# Comparing objects … (2/2)

- (Remember?) The `String` class has the method `int compareTo(Object)`, which returns:

  `0` → if the `String`s are equal;

  `<0` → if this object is less than the specified object;

  `>0` → if this object is greater than the specified object.

- The `Comparable` interface contains the `compareTo()` method.
  - If you wish to provide a natural ordering for your objects, you must implement the `Comparable` Interface.
  - Any object which is "comparable" can be compared to another object of the same type.
- There is only one method defined within this interface. So there is only one natural ordering for objects of a given type/class.

# Example using `Comparable` (1/2)

```
public class Employee implements Comparable<Employee> {
  int empID;
  String eName;
  double salary;
  static int i;

  public Employee() {
    empID = i++;
    eName = "unknown";
    salary = 0.0;
  }
  public Employee(String name, double sal) {
    empID = i++;
    eName = name;
    salary = sal;
  }
  public String toString() {
    return "EmpID  = " + empID + "\n" + "Ename  = " + eName + "\n" +
           "Salary = " + salary;
  }
```

```
(rest of the Employee class code)
  public int compareTo(Employee o1) {
    if (this.salary == o1.salary) return 0;
    else if (this.salary > o1.salary) return 1;
    else return -1;
  }
}
```

Aim: To order employees by salary value.

# Example using `Comparable` (2/2)

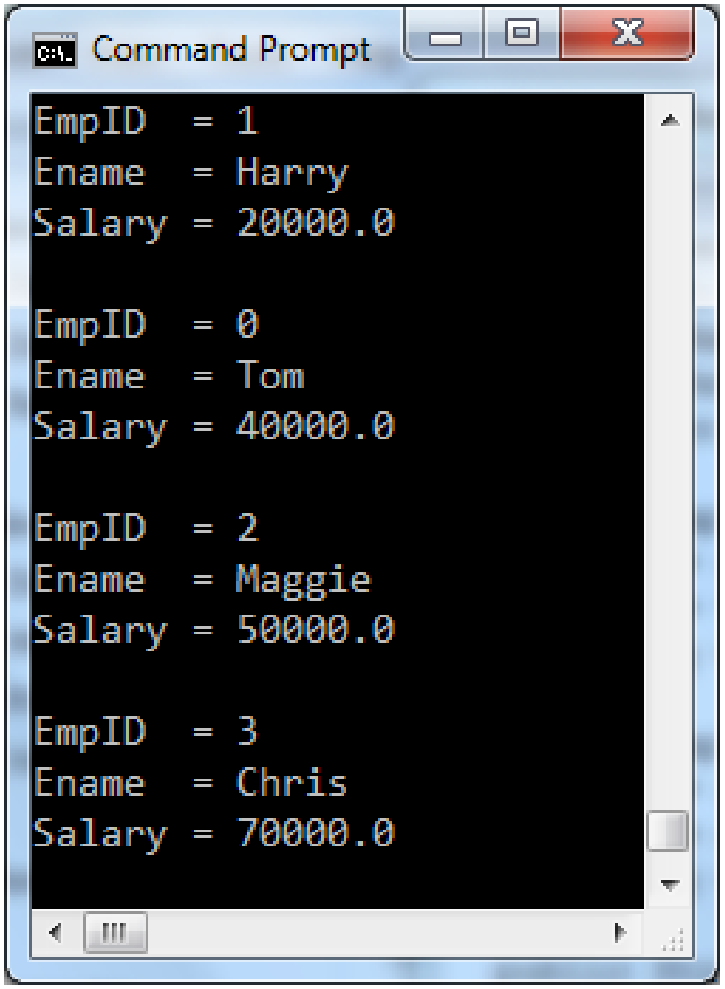
```
import java.util.*;
public class ComparableDemo {
  public static void main(String[] args) {
    List<Employee> ts1 = new ArrayList<Employee>();

    ts1.add(new Employee("Tom", 40000.00));
    ts1.add(new Employee("Harry", 20000.00));
    ts1.add(new Employee("Maggie", 50000.00));
    ts1.add(new Employee("Chris", 70000.00));

    Collections.sort(ts1);
    Iterator <Employee> itr = ts1.iterator();

    while(itr.hasNext()) {
      Object element = itr.next();
      System.out.println(element + "\n");
    }
  }
}
```

Output …



… and things for you to try out!